

Review article: dermatological complications of immunosuppressive and anti-TNF therapy in inflammatory bowel disease

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SUMMARY

Background

With the expanding list of medications available to treat patients with inflammatory bowel disease (IBD), it is important to recognise adverse events, including those involving the skin. Dermatological adverse events may be confused with extra-intestinal manifestations of IBD.

Aim

To review drug-related dermatological manifestations associated with immunosuppressive and anti-tumour necrosis factor (anti-TNF) therapy.

Methods

The literature was searched on PubMed for dermatological adverse events in IBD.

Results

Present thiopurine exposure was associated with a 5.9-fold [95% confidence interval (CI), 2.1–16.4] increased risk of developing non-melanoma skin cancer (NMSC). The peak incidence is highest in Caucasians over the age of 65 years with crude incidence rates of 4.0 and 5.7/1000 patient-years for present and previous use. In anti-TNF-exposed subjects, drug-induced lupus was reported in 1% of the cases and a psoriatic rash in up to 3% of the cases. Anti-TNF monotherapy increases the risk of NMSC ~2-fold to a rate of 0.5 cases per 1000 person-years. Cutaneous lymphomas have been rarely reported in subjects on thiopurine or anti-TNF drug monotherapy. Combination therapy seems to have an additive effect on the risk of developing NMSC and lymphoma.

Conclusions

Physicians need to be aware of the wide spectrum of dermatological complications of immunosuppressive and anti-TNF therapy in IBD, especially psoriasis and non-melanoma skin cancer. Vigilance and regular screening for non-melanoma skin cancer is recommended. Case discussions between gastroenterologists and dermatologists should be undertaken to best manage dermatological adverse events.

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INTRODUCTION

Crohn's disease (CD) and ulcerative colitis (UC) are chronic, complex relapsing and remitting inflammatory conditions associated with intestinal and extra-intestinal manifestations that cause significant patient morbidity,¹ as reflected by the cost in treating these diseases.² Over the decades, therapy has evolved from generic anti-inflammatory agents to specific targeted disease-modifying pharmacotherapy. The development of biological therapies, including anti-tumour necrosis factor alpha (TNF- α) monoclonal antibodies and anti-adhesion antibody molecules, has revolutionised the management of inflammatory bowel disease (IBD). Treatment paradigms have evolved to include earlier initiation of immune modulators and the more specific and effective anti-TNF α agents,³ although the optimal timing of their use is still an active point of discussion.

Broader use of these therapies has recently led to an increase in incidence of systemic side effects, such as dermatological manifestations. Adverse events involving the skin are now often seen in clinical practice. Recognising such manifestations is a key element of the required knowledge base of practitioners who manage patients with IBD, as these may be challenging at times to differentiate from extra-intestinal manifestations of IBD.

This article reviews the literature regarding dermatological complications of immune-modulating therapies specifically focusing on azathioprine, methotrexate, and antibodies to anti-TNF α and newer molecules as monotherapy or in combination.

AIMS AND METHODS

A Medline and Pubmed search was undertaken by two authors (G.M and A.L). The search criteria used were as follows: 'azathioprine' 'mercaptopurine' 'thiopurines' 'methotrexate' 'anti-TNF α ' 'infliximab' 'adalimumab' 'certolizumab' 'natalizumab' 'vedolizumab' 'ustekinumab' 'traficet-EN' 'tofacitinib'. The terms 'dermatology' 'cutaneous' 'side-effects' 'adverse events' 'complications' were used in the search query for each drug. After an initial title screen, all relevant articles were examined in full. Dermatological adverse events of each drug class will be discussed separately. Details on aetiopathology, incidence and treatment recommendations will be given where applicable. Results will be summarised in Table 1.

RESULTS

Thiopurines

The thiopurines, azathioprine (AZA) and mercaptopurine (MP) are commonly used as steroid-sparing

maintenance agents in patients with IBD. Azathioprine is the pro-drug that is converted to MP, which is then metabolised in the liver and gut tissue by hypoxanthine-guanine phosphoribosyltransferase into the active metabolites 6-thioguanine (thioguanine) nucleotides (6-TGN) (Figure 1). 6-TGN becomes incorporated into the replicating DNA and also inhibits *de novo* purine synthesis. This reduces the production of B lymphocytes, cytotoxic T-cell lymphocytes and plasma cells involved in pathogenetic mechanisms that drive inflammation in IBD. The major catabolic pathway involves xanthine oxidase and produces thiouric acid. A third pathway of metabolism is through thiopurine methyl transferase (TPMT), which produces several inactive metabolites.^{4, 5} A common polymorphism in the TPMT gene leads to a low activity variant allele in 10% of subjects. Decreased TPMT activity leads to shunting of MP metabolism to produce more 6-TG metabolites leading to myelotoxicity.^{6, 7} The majority of other adverse events, including dermatological manifestations, cannot be explained by altered TPMT metabolism.

Side effects to thiopurines can be broadly classified into dose-dependent non-allergic reactions and idiosyncratic dose-dependent presumed allergic reactions. Adverse reactions leading to intolerance occur in up to 30% of patients on thiopurines and may occur both at an early or late stage.⁸ It is thus helpful to subclassify adverse events to thiopurine in short-term, medium-term and long-term toxicities.

Short-term toxicity. Dermatological hypersensitivity reactions to thiopurines have been described in up to 12% of patients on thiopurines. The most common cutaneous manifestations include urticarial or a maculopapular rash.⁹

Sweet syndrome (SS) is an abrupt eruption of painful erythematous plaques or nodules, often in the setting of fever. This has been described in 10 patients all within 2 weeks of starting azathioprine.⁹ It is important to note that SS has been described as a cutaneous extra-intestinal manifestation of IBD itself. Similar to SS, erythema nodosum and sterile pustules have been described in the setting of thiopurines,¹⁰ although these could be a reflection of the underlying disease rather than an adverse event. There have also been case reports of acute generalised exanthematous pustulosis described as diffuse, small nonfollicular pustules associated with fever, neutrophilia and spontaneous resolution within 15 days.¹¹ In all the cases, resolution of the symptoms occurred after cessation of therapy. Erythema gyratum

Table 1 | Summary of the most common dermatological adverse events of IBD-related immune therapy. Specific disease details and recommendations are included

Drug	Adverse event	Recommendation
Thiopurines	Drug hypersensitivity	Occurs in 12% of the cases. Prompt cessation of therapy is usually followed by complete resolution of the manifestation. Sixty per cent of patients would be tolerant to the other thiopurine.
	Sweet's syndrome	Prompt cessation of therapy is usually followed by complete resolution of the manifestation
	Varicella zoster	Recommended to vaccinate naïve patients prior to starting therapy. Withdraw therapy temporarily and treat with oral or intravenous anti-viral, depending on the severity of the presentation.
	Non-melanomatous skin cancer (NMSC)	Peak incidence in Caucasians over the age of 65 years (5.7/1000 patient-years). Drug withdrawal is not routinely recommended. Each case should be discussed individually in a multidisciplinary meeting. Patient education, skin protection and routine yearly skin checks recommended.
	Cutaneous lymphoma	Drug withdrawal usually recommended. Incident rate is 0.90 per 1000 patient-years with an odds ratio of 5.28.
Methotrexate	Common reactions	These include a generalised maculopapular rash (15%), alopecia (8%), photosensitivity (5%) and urticaria (4%). Drug withdrawal not routinely recommended.
Calcineurin inhibitors	Common reactions	Skin infections occur in 7% of the cases. Hirsutism, acne and gingival hyperplasia occur in more than 3% of the cases, while drug hypersensitivity and brittle finger nails happen in less than 2% of the cases. Drug withdrawal not routinely recommended.
Anti-TNF α	Drug hypersensitivity	Incidence of 1%. Prompt drug withdrawal and an in-class switch to another agent suggested.
	Skin infections	Incidence of up to 0.5%. Temporary cessation of drug suggested until clinical improvement observed.
	Drug-induced lupus erythematosus	Incidence of ~1%. Positive antibody titres in the absence of symptoms are not an indication for drug cessation. 94% of the cases respond to drug cessation. Re-challenging with another drug is a potential option with a low recurrence observed.
	Psoriasis	Occurs in 3% of treated cases and commonly in women. Changes to a guttate or pustular phenotype. Prompt drug withdrawal is not recommended. If the psoriasis fails to improve upon topical, systemic or phototherapy, then the biological agent should be withdrawn. Drug re-challenge is a possible option, although associated with a high recurrence (~50%).
	Eczema	Anti-TNF α therapy should only be withdrawn in the event of failure of topical therapy
	IGD	An indication for prompt cessation of biological therapy
	NMSC	A reported prevalence of 0.3–1.4%. Recent and persistent biologic use is associated with a ~2-fold increase in the incidence of NMSC. Patient education, skin protection and yearly skin checks recommended. Prompt drug withdrawal is not routinely recommended.
Combination therapy	Opportunistic infections	3- to 4-fold increase in incidence. Risk is increased in patients over the age of 50 years and receiving 2 or more immune modulators. Varicella zoster vaccination is suggested in naïve patients prior to embarking on such therapy.
	Skin cancer	Past use of combination therapy is associated with 5.85-fold increased risk of NMSC as compared to 3.56-fold risk in patients previously exposed to thiopurine therapy and 2.07-fold increased risk in patients on anti-TNF α monotherapy. Current use of combination therapy is associated with 6.75-fold increased risk of NMSC as compared to 4.27-fold risk ascribed to thiopurine therapy and 2.18-fold increased risk in patients presently on anti-TNF α monotherapy. Patient education, skin protection and yearly skin checks recommended. Prompt drug withdrawal is not routinely recommended. Skin lymphoma has been described in the setting of IBD treated with immune therapy. A standard incidence ratio (SIR) of 3.23 is ascribed to subjects on combination therapy when compared with naïve patients. Patients on anti-TNF α monotherapy had a SIR of 1.7 when compared with other patients on thiopurine monotherapy. Lymphoma is usually an indication for prompt cessation of therapy.

Table 1 (Continued)		
Drug	Adverse event	Recommendation
Natalizumab	Hypersensitivity	Allergic hypersensitivity described in up to 46% of the cases.
Ustekinumab	Psoriasis	Worsening of psoriasis has been seen in 4 cases after exposure to ustekinumab.
	Herpes zoster	Use of this drug is felt to be a risk factor for herpes zoster infection with the high dose (90 mg) associated with a 3-fold increased incidence of this infection when compared with patients treated with a lower dose (45 mg).
Tofacitinib	Skin rashes	Acne in 13.2–16.8% of the cases and a nonspecific rash in up to 11.3% of exposed cases.

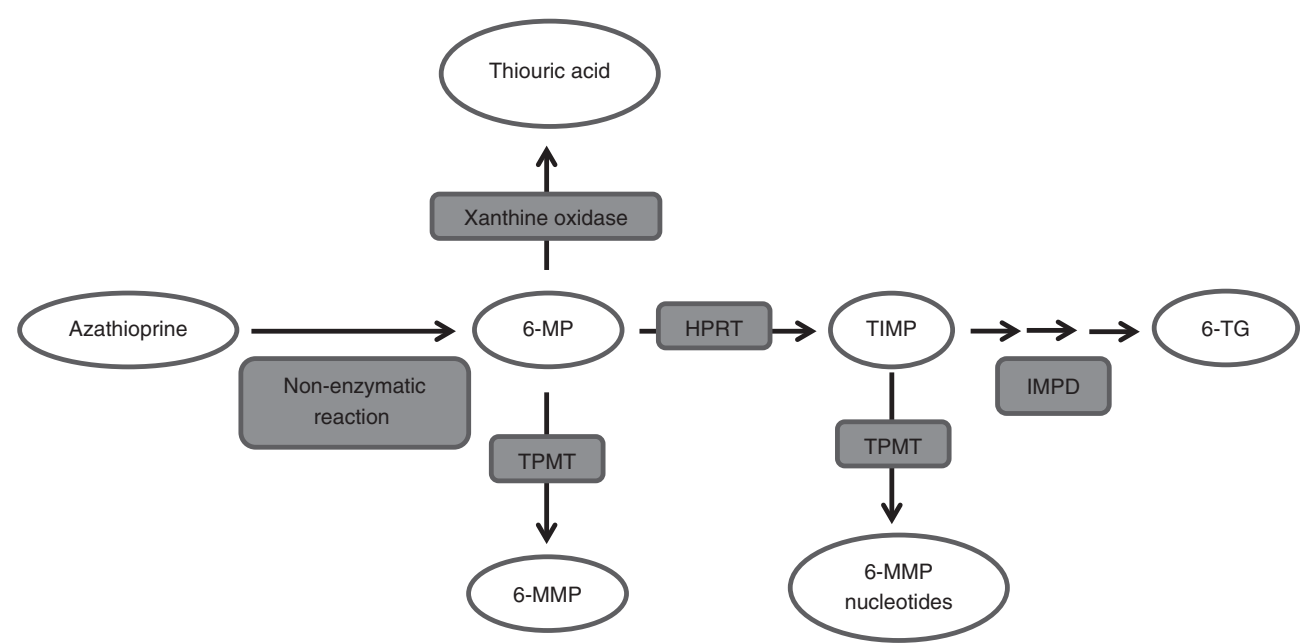


Figure 1 | Simplified graphic representation of the metabolism of azathioprine into its active metabolite 6-tioguanine (6-TG). MP, mercaptopurine; TPMT, thiopurine methyltransferase; HPRT, hypoxanthine-guanine phosphoribosyltransferase; 6-MMP, 6-methylmercaptopurine; TIMP, thioinosine monophosphate; IMPD, inosine monophosphate dehydrogenase.

repens has been ascribed to azathioprine as well.¹² Confirming hypersensitivity with a drug re-challenge can be potentially dangerous, as this may induce worse symptoms. This should be carried out in a controlled setting. Sixty per cent of subjects allergic to azathioprine are tolerant to mercaptopurine, indicating that the immunological hypersensitivity is directed to the imidazole rather than the thiopurine moiety of azathioprine.^{8, 13, 14}

Medium-term toxicity. It is possible that azathioprine increases the infection risk in the absence of neutropenia, although the evidence is limited. Organ transplant recipients receiving azathioprine in conjunction with other immune suppression do have an increased infection risk,

although this might be due to the level of immune suppression rather than due to the drug *per se*. Varicella zoster (VZV) infections have been shown to be commoner in IBD patients treated with azathioprine,¹⁵ although IBD subjects treated with thiopurine monotherapy have not been shown to have an overall increased infection risk.¹⁶ It is generally recommended to vaccinate patients for VZV prior to starting thiopurine therapy. At the onset of infection with VZV in thiopurine-exposed subjects, it is recommended to withdraw therapy and institute oral or intravenous anti-viral therapy depending on the severity of infection.¹⁷

Alopecia has been reported in medium- to long-term exposure, although its aetiology is still unclear.¹⁸ Significant alopecia may lead to discontinuation of therapy.

Long-term toxicity. Historically, immunosuppressive therapy with thiopurines has been associated with cutaneous malignancies in solid organ transplant patients,^{19–24} as early as in the 1970s.^{25, 26} This observation is especially true in renal transplant recipients receiving thiopurine immunotherapy. The level of suppression of the immune system aimed for in IBD is however considerably less than in transplant recipients.

Skin cancer: Patients receiving higher doses for extended periods of time have been shown to have a 100-fold increased risk for non-melanoma skin cancers (NMSC), specifically squamous cell carcinoma when compared with the normal population.^{27, 28} Similar findings, although not to the same extent, have been noted in the rheumatology literature.^{29–31} This may be the result of an attenuation of immune surveillance secondary to *in situ* depletion of antigen-presenting Langerhans cells and dermal CD4⁺ and CD8⁺ subsets.³² Moreover, cutaneous accumulation of 6-TGN could be potentially carcinogenic under relevant exposure to ultraviolet (UV) A.³³

Recent progress has been made in understanding the mechanism of photo-oncogenesis in UV-exposed, thiopurine-treated subjects. UVA (wavelength 320–440 nm), which constitutes 95% of solar radiation, is poorly absorbed by purines in DNA and is thus considered harmless. However, 6-TG has a maximum absorbance at 340 nm, and absorption of UVA photons by 6-TG-substituted DNA creates oxygen radical species, which can induce DNA mutagenesis and eventually NMSC.

In IBD, most of the data available up to now are retrospective and limited to case series.^{34–39} An increased incidence of skin squamous cell carcinomas was shown among 1961 IBD patients.⁴⁰ In a Danish cohort of 5546 UC patients, the relative risk of NMSC was noted to be 1.4 [95% confidence interval (CI), 1.0–1.9] when compared to the general population.⁴¹ In a population-based cohort of 4776 CD patients, the risk of squamous cell skin cancers was described as 5.5 (CI: 2.0–11.9).⁴² None of these studies controlled for treatment. Few studies have specifically examined the influence of immune suppression on cancer risk in patients with IBD, with smaller studies failing to identify a correlation.⁴³ A large retrospective study, comparing the incidence of NMSC from computerised records of 53 377 IBD patients and matched controls, has suggested a positive association.³⁷ This study demonstrated a risk of NMSC in patients who had been exposed to biological and thiopurine ther-

apy for 90 days. A further increase in risk was observed in patients exposed to thiopurines for more than 12 months. Latitude was described as a significant variable, but the confounding effect of ethnicity was not described. A recent retrospective South African study described similar findings.⁴⁴

A large prospective study specifically looking at the incidence of NMSC in IBD in the CESAME French nationwide observational cohort has provided some definitive data.⁴⁵ The CESAME cohort was designed to assess any cancer risk in IBD patients. A cohort of 19 486 consecutive IBD patients was enrolled from May 2004 to June 2005. They were followed up until 31st December 2007 leading to 49 719 patient-years of follow-up. The highest incidence of NMSC was observed in Caucasians over the age of 65 years, with the crude incidence ratios being 0.84/1000 patient-years in the patient group naive to azathioprine and then rising to 4.04 and 5.7/1000 patient-years in patients with current and previous exposure respectively. A multivariate Cox regression model identified that present thiopurine treatment [hazard ratio (HR): 5.9, CI: 2.1–16.4] and past thiopurine exposure (HR: 3.9, CI: 1.3–12.1) were significant risk factors for NMSC.

The available data on the incidence of melanoma skin cancers in thiopurine-exposed patients are very limited, with only three case reports available in the literature. In one case, a renal transplant recipient developed metastatic malignant melanoma after three decades of immune suppression with azathioprine and prednisolone⁴⁶ and in the other two cases, metastatic malignant melanoma presented after a decade of azathioprine exposure.⁴⁷

Lymphoma: Cutaneous T-cell lymphoma in the setting of previous Epstein–Barr virus (EBV) exposure has also been recently described in a 67-year-old patient exposed to prednisolone and azathioprine for dermatomyositis for 2 years. The skin ulcers promptly regressed on withdrawing the immune suppression, with the patient reported to be asymptomatic until follow-up. The pathological link between EBV infection and lymphoma risk could be explained by the fact that thiopurines induce a defective T-cell-mediated response to EBV-infected B-cells, resulting in a malignant clone of B-cells. Moreover, the EBV-encoded latent proteins seem to possess oncogenic properties as well.⁴⁸ The effect of thiopurine therapy on the incidence of lymphoma in IBD has been investigated in several reports. Three meta-analyses have shown conflicting results, with one study showing a

4.18-fold (CI: 2.07–7.51) increased risk of developing lymphoma in IBD patients exposed to thiopurines. This study was later criticised, as normal subjects were recruited in the comparative group, thus not allowing for the IBD-related lymphoma risk.⁴⁹ In a separate meta-analysis, this risk was found to be much more modest (1.69-fold).⁵⁰ In a later meta-analysis, the authors⁵¹ failed to find an increased risk, although in this study, IBD patients naïve to immune suppression rather than a normal control group were used as a comparative group. Two later population studies once again showed conflicting results. In one study, information on IBD and relevant medications was obtained from an IBD Registry spanning between 1996 and 2009.⁵² Lymphoma cases were ascertained from a cancer registry. Lymphoma incidence was compared between the IBD cohort and the general population. IBD patients naïve to immune-modulator therapy and anti-TNF α medication were shown not to have an increased lymphoma risk [standardised incidence risk ratio (SIRR), 1.0, CI: 0.96–1.1]. Present use of thiopurine therapy was shown to give a slightly increased risk of lymphoma (SIRR: 1.4, CI: 1.2–2.7), with the risk significantly decreasing upon stopping therapy (SIRR for past exposure: 0.3, CI: 0.2–0.4). The Dutch nationwide study failed to show an increased risk.⁵³

A large prospective study specifically looking at the incidence of lymphoproliferative disorders in thiopurine-exposed IBD subjects in the CESAME French nationwide observational cohort has provided some definitive data.⁵⁴ A total of 19 486 patients with IBD were enrolled in a nationwide French cohort with reported details of immunosuppressive therapy, cancer incidence and mortality during the observation period. The incidence rates of lymphoproliferative disorder were 0.90 per 1000 (CI: 0.50–1.49) patient-years in those on current therapy, 0.20/1000 (0.02–0.72) patient-years in those on previous therapy and 0.26/1000 (0.10–0.57) patient-years in those who were naïve to thiopurines ($P = 0.0054$). The multivariate-adjusted hazard risk ratio of lymphoproliferative disorder between patients receiving thiopurines and those who had never received the drugs was 5.28 (2.01–13.9). It is important to note that all these studies were on patients with noncutaneous lymphomas. With no data being available on the specific incidence risk of cutaneous lymphomas, these data might not be an exact reflection of the risk posed to patients.

It seems rational that baseline and routine comprehensive, full-body dermatological examinations should be advocated. Advice on photoprotection should be sug-

gested by practitioners. Patients should have a very low threshold to report any suspicious lesions to dermatologists with a special interest in skin cancer. Withdrawal of the thiopurine therapy is not mandatory, but in some instances preferable after discussion with the IBD physician, when other less immune-suppressive therapy might be available.⁵⁵ Persistence of thiopurine therapy in the setting of NMSC should be accompanied by strict and frequent dermatology follow-up.

Methotrexate

Methotrexate is a structural analogue and anti-metabolite that inhibits folic acid metabolism by blocking the conversion of dihydrofolic acid to folinic acid through the inhibition of dihydrofolate reductase.⁵⁶ It leads to inhibition of enzymes involved in purine metabolism, leading to adenosine accumulation, inhibition of T-cell activation and the suppression of intercellular adhesion molecule expression by T-cells.⁵⁷ Side effects to methotrexate are often dose-related and include nausea, hepatitis, bone marrow suppression and cutaneous manifestations.⁵⁸

Common dermatological reactions include a generalised maculopapular rash (15%), alopecia (8%), photosensitivity (5%) and urticaria (4%).⁵⁹ Dose-dependent cutaneous manifestations of methotrexate occur in the setting of cytotoxic T-cell-induced and mononuclear cell-induced apoptosis and tend to affect areas of rapidly dividing keratinocytes, such as the oral mucosa.⁶⁰ These occur as erosions, ulcers, haemorrhagic bullae or necrotic areas involving the oral and vaginal mucosae and skin on the knees and back.⁶¹ Other serious dermatological syndromes such as Stevens–Johnson, toxic epidermal necrolysis and erythema gyratum repens have also been documented as a result of the apoptotic mechanisms of methotrexate.^{62–64}

Methotrexate-associated lymphoproliferative disorders are rare, with more than 100 cases being described in the literature.⁶⁵ Forty per cent of these have extra-nodal involvement. Ten cases of methotrexate-associated primary cutaneous lymphomas have been described.^{66, 67} The majority of these tumours are of B-cell origin with half being EBV-positive. These lymphomas are multifocal and tend to regress on withdrawing the methotrexate. This strict relationship was highlighted more strongly in a case of an EBV-positive cutaneous B-cell lymphoma occurring specifically at the site of methotrexate injection.⁶⁸ The lesion promptly regressed on withdrawing the offending drug. Local immunosuppression leading to tumour proliferation is probably the aetiology in this case.

A 3-fold increased risk of malignant melanoma has been described in methotrexate-exposed rheumatology patients.⁶⁹ A number of case reports have underlined this association.^{65–68} As no data are available in the IBD literature, it could be potentially hazardous to extrapolate this risk from rheumatology to IBD subjects. Despite this, physicians should be aware of the increased risk of malignant melanoma in methotrexate-exposed IBD subjects.

Calcineurin inhibitors

The mechanism of action of calcineurin inhibitors is thought to result from inhibition of nuclear translocation of the transcription factor NFAT (nuclear factor of activated T-cells), thus downregulating T-cell-related cytokines.⁷⁰

Ciclosporin is effective in severe acute UC^{71, 72} and it has a role in steroid-resistant cases⁷³ as well. Due to its limited therapeutic window and extensive side effect profile, its use has only been very limited.

From a dermatological standpoint, ciclosporin has a limited side effect profile. Based on historical experience in renal, liver and heart transplant patients, skin and wound infections were the commonest dermatological side effects with an incidence of 7%.⁷⁴ Hirsutism, acne and gingival hyperplasia occurred in more than 3% of the cases, while drug hypersensitivity and brittle finger nails were observed in less than 2% of the cases. Brittle hair, oral ulcers and pruritus occurred rarely.⁷⁴

Cutaneous malignancies have long been associated with the chronic ciclosporin use, with the highest incidence of cancer being observed in patients with psoriasis who were previously exposed to methotrexate or photochemotherapy.⁷⁵

Presently, 29 cases of cutaneous T-cell lymphoma in a post-transplant setting have been described.^{76–80} These have occurred predominantly in male patients who have been previously exposed to immunosuppressants (typically calcineurin inhibitors) and have a later onset than their B-cell counterparts. Due to the fact that ciclosporin and another calcineurin inhibitor, tacrolimus, are usually used as bridging therapy to thiopurines in patients with acute UC, the treatment duration does not usually last longer than 6 months.⁸¹ To this effect, calcineurin-associated lymphoproliferative disorders in IBD are not commonly seen.

Anti-tumour necrosis factor alpha (TNF- α)

Mechanism of action. Tumour necrosis factor- α (TNF α) is a pro-inflammatory cytokine involved in the pathogen-

esis of immune-regulated conditions, such as IBD. Over the last decade, the development of anti-TNF α agents has revolutionised the treatment of IBD. These biological treatments include infliximab, a chimeric IgG1 monoclonal antibody, adalimumab, a recombinant monoclonal antibody and certolizumab, a PEGylated Fab fragment of a humanised anti-TNF α monoclonal antibody. These peptides bind and inhibit the action of TNF α . Both infliximab and adalimumab have been shown to be effective therapies in CD^{82–84} and UC.^{85, 86}

To date, several randomised controlled trials have shown that early treatment with anti-TNF α therapy improves outcomes.^{3, 87, 88} Inducing early remission and mucosal healing improves long-term outcomes.^{89–91} As a result of these data, the use of anti-TNF α medication in conjunction with anti-metabolite therapy like thiopurines and methotrexate will be more widespread in the forthcoming years.

Dermatological side effects have been described. A prospective study reported an association between TNF α antagonists and a variety of dermatological side effects in 150 rheumatological patients. In this cohort, 35 patients experienced cutaneous infections, chronic inflammatory skin diseases or other skin disorders; thus indicating that a fifth of patients exposed to anti-TNF α therapy are at risk of potential dermatological side effects.⁹² This study was hampered by its small size. Another study exploring adverse cutaneous effects in rheumatological patients on anti-TNF α agents indicated that 12.7% of the study population developed immune-mediated skin lesions.⁹³ Relevant skin complaints described in anti-TNF α -treated patients will be described separately.

Local skin complications and infections. Local complications at the site of drug administration have been frequently reported. From the German and the British Society of Rheumatology Biologics registers, skin infections are second only to respiratory infections in biologics-exposed patients, with 1.5% of patients suffering from dermatological infections.⁹⁴

Complications such as cellulitis (0.5% vs. 0% control) and skin ulcers (0.1% vs. 0% control) are common examples.⁹⁵ Over a prospective follow-up period of 3.25 years, the same study showed that 1% of patients experienced an acute infusion reaction (defined by adverse event occurring during the infusion or within 2 hours of terminating the infusion) at the injection site, manifesting with pruritus or urticaria.

In a descriptive study looking at skin complications following anti-TNF α exposure, 13 of 35 patients presented

with a new onset of an infectious skin complications during anti-TNF α treatment.⁹² These include herpes simplex virus (HSV) reactivation (1/13), microbial eczema (5/13), acute cutaneous staphylococcal infection (1/13), *tinea corporis* (1/13) and *pityriasis versicolor* from the yeast *Pityrosporum ovale* (3/13).⁹² The authors did not find a correlation between a particular anti-TNF α agent and any specific infectious process, thus indicating that these infections are a class effect.

Moreover, infliximab use has been associated with the development of acute folliculitis or superficial granuloma⁹⁶ and acute pustular dermatitis accompanied by necrotising fasciitis.⁹⁷

Drug-induced systemic lupus erythematosus (SLE) – DILE. The aetiology of anti-nuclear antibody development and lupus-like syndrome in patients undergoing TNF α therapy is not completely understood. Anti-TNF α -induced leucocyte apoptosis may release nuclear antigens, thus inducing antibody formation and phenotype positivity.⁹⁴ Systemic modulation of TNF α expression may inhibit apoptosis, leading to accumulation of antigenic nuclear debris.^{98–100} Anti-TNF α -induced inhibition of cytotoxic cells may reduce the clearance of antibody producing B-cells.¹⁰¹ The ‘cytokine-shift’ hypothesis proposes that anti-TNF α drug exposure induces a Th1 to Th2 shift in cytokine expression, leading to a change in downstream signalling and induction of a serological lupus-like phenotype.^{102–105}

Autoantibody seropositivity should be distinguished from DILE. Induction of autoantibodies has been described in clinical trials using infliximab in rheumatoid arthritis.^{106–108} Pooled analysis of these placebo-controlled and open-label randomisation studies shows that the incidence of anti-nuclear antibody (ANA) positivity increases from 29% pre-treatment to 53% after infliximab exposure.¹⁰⁹ Such a phenomenon has been observed in early studies in CD with anti-DNA seropositivity described after 10 weeks of treatment.^{110–112} Half of the infliximab-treated patients in clinical trials developed positive ANA ($n = 5707$), but 20% of placebo-exposed patients also demonstrated seropositivity ($n = 1600$).¹¹³ Four per cent of certolizumab pegol patients, 10–12% of adalimumab patients, 15% of etanercept patients and 20% of infliximab patients (compared with none of placebo controls) develop anti-double stranded (ds) DNA antibodies after exposure to these drugs. Antibody-positive patients develop IgM subtype antibodies rather than the more pathogenic IgG.^{113, 114} Due to the aetiology of this manifestation, DILE is not a

drug-specific phenomenon, but rather a class-specific effect.

DILE is an uncommon phenomenon, with most cases reported by rheumatological literature (Figure 2). DILE-IBD-specific literature is scarce, mostly limited to case reports and small case series. Predicting DILE behaviour in an IBD-specific context based on data from the rheumatology literature might be inaccurate due to the different pharmacokinetic nature of these two autoimmune diseases.

The BIOGEAS, a Spanish registry on the use of anti-TNF α therapy in patients with autoimmune diseases, has reported 105 DILE cases between January 1990 and May 2008 (45 infliximab, 35 etanercept and 22 adalimumab).¹¹⁵ Only 10 of these cases were in patients with CD. Other case series have reported similar numbers.^{116, 117} After reviewing Food and Drug Administration of the United States indications and drug-specific labels, an incidence of approximately 1% is postulated.^{113–115, 118–120} Clinical features are mostly cutaneous (a malar rash is the most common presentation in over a third of the cases).

Photosensitive and discoid rashes, oral ulcers, arthralgias, serositis and haematological abnormalities have all been observed.^{115–117} Lupus nephritis has never been described in the setting of DILE. Comparatively, DILE due to anti-TNF α drug exposure leads to more cutaneous symptoms.^{115, 121, 122}

Another noticeable difference is the pattern of antibody positivity. Classical DILE leads to a high incidence of ANA (>99%) positivity and anti-histone positivity (>95%). Anti-dsDNA antibody is rarely positive (1%). In



Figure 2 | A close-up of the subacute cutaneous lupus rash (drug-induced systemic lupus erythematosus - DILE), related to exposure to anti-TNF therapy.

DILE, due to anti-TNF α drug exposure, patients are usually anti-dsDNA-positive (90%) and anti-histone-positive in only 57% of the cases. Positive extractable nuclear antigens and hypocomplementaemia are more routinely seen in DILE due to anti-TNF α drugs.^{121, 122}

There is currently no recommendation on the prevention of DILE in anti-TNF α -exposed patients. Antibody positivity in the absence of clinical symptoms is not a reason to stop the biological therapy. In 94% of DILE cases, this syndrome reverses on withdrawal of the offending drug. BIOGEAS report that 40% of the patients who suffered DILE needed corticosteroid treatment, with 12% needing additional immunosuppressive therapy.¹¹⁵

A recent report suggested that re-challenging patient with a second anti-TNF α drug is a potential option. Eight of the thirteen patients reported were re-challenged with a second anti-TNF α drug and only two showed a recurrence of their symptoms.¹¹⁷

Psoriasis. Psoriasis is a noncontagious, inflammatory systemic disorder with numerous cutaneous variants or manifestations (Figure 3). In up to 10% of cases, it has been associated with inflammatory conditions, such as IBD.¹²³ Both infliximab and adalimumab have been shown to be an effective therapeutic option for patients with moderate-to-severe psoriasis.^{124, 125} Paradoxically, anti-TNF α treatment has been implicated in the *de novo*



Figure 3 | Showing evidence of plaque psoriasis on the back (a) and lower limbs (b). Close-up picture shown in Figure 3c. The patient had been exposed to an anti-TNF biologic for her refractory Crohn's disease. Cessation of the anti-TNF therapy resulted in complete resolution of this manifestation, as seen in Figure 1d.

development of psoriasis and this has been reported with infliximab,^{126, 127} adalimumab^{127, 128} and certolizumab pegol.^{129, 130}

Since the first reported case of psoriasiform plaques in patients with CD treated with infliximab in 2004,¹²⁶ the overall rate of anti-TNF α -associated psoriatic-like manifestations has been deemed to range from 1 case per 1000 years treated¹³¹ to an overall frequency of 3%.¹³² Anti-TNF α -induced psoriasis is more common in women (70%).¹³³ The typical time lag from initiation of therapy to clinical manifestation of the disease ranges from 2 to 6 months.^{131, 134} Half of the patients examined in one study had exacerbations of previously diagnosed psoriatic conditions.⁹² However, some patients develop this adverse effect, despite no personal or family history of the disease.

For the most part, plaque guttate and pustular psoriasis have been documented. However, palmoplantar pustular disease appears to be more common than in idiopathic psoriasis,¹³⁵ as is specifically the case in *de novo* psoriasis described in patients treated with certolizumab pegol.^{129, 130}

In one study, the authors report that 8 of 35 patients on an anti-TNF α agent showed psoriasis-like skin lesions: 3 had psoriasis vulgaris (2 of whom had relapse of previously diagnosed disease) and 3 had palmoplantar pustular psoriasis.⁹² A scalp distribution of this psoriasiform manifestation has been recently shown to be as common as a palmoplantar distribution.¹³³ A substantial number of biopsies were gathered in many of the cases reported that have shown the typical histological features of psoriasis. In some cases, atypical findings have been documented that represent a psoriasiform drug reaction.¹³⁶ This is associated with a prompt resolution of psoriasiform symptoms after drug cessation and a relapse after re-challenging.

An important differential diagnosis is psoriasiform lesions reactive to remote infections (pseudopsoriasis).¹³⁷ Interestingly, three patients with suspected drug-associated psoriasis were ultimately found to have *Chlamydia trachomatis*, leading to keratoderma blenorrhagicum.¹³⁸ Anti-TNF α drug-induced psoriasis occurs through the activation of autoreactive T-cells¹³⁹ and through interferon-alpha (IFN α) activity.¹⁴⁰ This is explained by the complex interplay between TNF α and IFN α . The main source of IFN α is in the plasmacytoid dendritic cells that have been described in early psoriatic skin lesions.^{141, 142} T-cells are thereafter recruited through IFN α -mediated upregulation of CXCR3 on T-cells. IFN α also activates the secretion of pro-inflammatory cytokines by myeloid

dendritic cells, specifically interleukin (IL)-12 from activated dendritic cells and IL-23 from keratinocytes.¹⁴³

TNF α plays an immune-regulatory role by decreasing IFN α expression via inhibition of maturation of plasmacytoid dendritic cells.¹⁴⁰ Thus, anti-TNF α agents result in increased IFN α levels, potentially explaining the onset of psoriasis (Figure 4). Studies in anti-TNF α -associated psoriasis show greater upregulation of IFN α within skin plaques than in idiopathic disease.¹³⁵ This proposed molecular theory does provide a valid explanation for the development of psoriasis. However, it does not explain the variable duration between treatment initiation and disease onset.

An evaluation by a dermatologist is suggested to properly diagnose psoriasis and to confidently exclude any relevant differential diagnoses. Topical therapy without prompt cessation of the anti-TNF α drug is usually suggested in mild-to-moderate cases (<than 5% of skin affected). Initial therapy with topical corticosteroids or other agents (keratolytics, and vitamin D analogues),

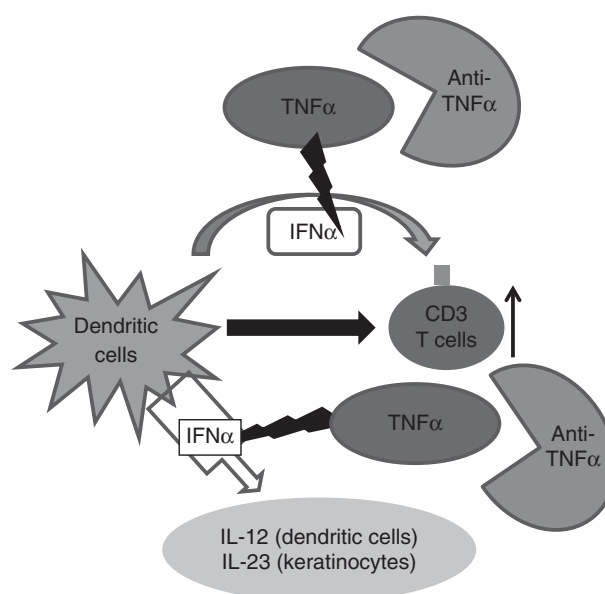


Figure 4 | Diagrammatic representation of the complex interaction between IFN α and TNF α . The major source of IFN α in the skin is the dendritic cells. This leads to an increase in CD3 T-cell recruitment and an increase in IL-12 and IL-23 by dendritic cells and keratinocytes respectively. These pathways are thought to be important in the aetiopathology of psoriasis. TNF α decreases the expression of IFN α by dendritic cells. Inhibition of TNF α , with monoclonal antibodies could potentially induce or worsen pre-existing psoriasis.

systemic therapy (methotrexate, retinoids or ciclosporin) or phototherapy seems to be effective in most cases.^{144–146} If no improvement is noticed, then the physician, in conjunction with the patient, should conduct a risk–benefit analysis. Stopping the anti-TNF α drug should be considered. Using a second anti-TNF α drug is an option, but a high recurrence (52%) or persistence of symptoms has been reported.^{133, 147}

Alopecia. Alopecia has to be differentiated from psoriasis as, often, these two manifestations may co-exist.¹⁴⁸ Adalimumab-induced psoriasis of the scalp with diffuse alopecia has been reported.^{93, 149, 150} In one case, the patient developed diffuse alopecia and characteristic psoriatic plaques on the scalp, trunk and extremities, 3 months after starting therapy with adalimumab following loss of response to infliximab. Cessation of therapy resulted in resolution of the scalp lesions and complete hair re-growth.¹⁴⁹ Other cases include a 39-year-old woman with androgenetic alopecia and telogen hair loss 8 months after commencing therapy; and a 23-year-old woman with alopecia areata immediately after therapy initiation, and again with repeat exposure to the medication 1 year later.¹⁵⁰

A slightly more aggressive case of alopecia has been described in a young female, with CD, after initiating monotherapy with adalimumab. The patient then proceeded to develop scalp psoriasis with progressive alopecia. Biopsy specimens were confirmatory of both conditions.¹⁴⁹ Prompt investigation and identification of the causes of alopecia are of paramount importance. Certain variants, like scarring alopecia, as in this case, are irreversible, despite cessation of the offending agent.

Lichenoid reactions. Lichenoid drug reactions have also been described in the literature in rheumatological and gastroenterological patients treated with infliximab,^{96, 151–155} etanercept^{153, 156, 157} and adalimumab.^{93, 152, 153, 158} The spectrum of histologically similar lichenoid reactions can be categorised into three main groups:¹ those with clinical features similar to lichen planus (LP), ² a subgroup with nonspecific macular or papular morphology and ³ psoriasiform eruptions with lichenoid histology. In all documented TNF α -related LP-like eruptions, lichenoid interface dermatitis was confirmed by histology.

Cutaneous lichen planus is a pruritic, violaceous, flat-topped polygonal papule that can coalesce to form plaques. For the most part, the term lichen planus-like eruptions can be used to describe eruptions that are

clinically similar to idiopathic lichen planus. Descriptors for this condition have included the following: violaceous papular lesions with mild scaling;¹⁵⁴ erythematous polygonal papules;¹⁵¹ purple-coloured erythematous plaques;¹⁵⁷ reticulate white mucosal plaques;^{155, 158} maculopapular exanthema;¹⁵³ and lichenoid interface dermatitis based on histology.^{126, 136} The pathogenesis of this condition relates to the release of antigens at the basal cell layer and dermo–epidermal junction. These are thought to elicit a cell-mediated response that results in infiltration of T lymphocytes and hyperplasia of Langerhans cells. Possibly, lichen planus can be mediated by dendritic cells and T-cells with TNF α playing a key role. Paradoxically, reports have indicated that anti-TNF α drugs may be used as therapy in some severe cases of lichen planus.¹⁵⁹

Eczema. Another documented side effect is eczema, either as atopic dermatitis-like eruptions or as eczematous skin lesions. The development of eczema is related to altered immune regulation. Eczema is felt to be associated with deficiency and malfunction of circulating CD8+ suppressor/cytotoxic T-cells and reduced activity of natural killer cells after anti-TNF α therapy exposure. This immunological dysregulation involves an alteration in Langerhans cells (LC), monocytes, macrophages, lymphocytes, eosinophils, mast cells, basophils and keratinocyte expression.

Lee *et al.* reported that 8 of 16 patients developed eczema predominantly in the antecubital and popliteal fossae, and also on the face or trunk.⁹² Moreover, these eczematous lesions have been described as a non-allergic reaction.¹⁶⁰ The anti-TNF α therapy should be withdrawn only in cases where topical therapy is unsuccessful. In all cases, an allergic reaction to the offending drug should first be excluded.

Vitiligo. Vitiligo is a common skin disorder that results in loss of melanocytes leading to clinically apparent irregular cutaneous pigmentation. It is associated with an underlying immunological dysregulation that results in circulating autoantibodies against melanocytes. Experimental evidence shows that TNF α may play a role in the pathogenesis of nonsegmental vitiligo. Subjects showing high cytoplasmic expression of TNF α are characterised by higher vitiligo disease activity scores. Interestingly, infliximab has been used with some success in the treatment of vitiligo.¹⁶¹ Paradoxically, in some patients, infliximab use has been shown to worsen vitiligo¹⁶² and, in a follow-up study of 252 rheumatological patients,

one patient was described as having developed vitiligo *de novo* after infliximab exposure.⁹³ Vitiligo can be a potential dermatological complication to biological therapy, although its actual pathogenesis is as yet uncharacterised.

Acneiform eruptions. Acne vulgaris is a common dermatological condition that is associated with a number of pharmaceutical agents, including azathioprine, androgens, corticosteroids, ciclosporin and many more. There have been only five case reports of acneiform eruptions associated with infliximab ($n = 3$) and adalimumab ($n = 2$) over the past number of years.^{163, 164} The immunopathological rationale behind this development remains unknown.

Vasculitis. The occurrence of vasculitis with anti-TNF α agents is well reported.^{93, 96, 153, 165} In the rheumatological literature, the cumulative incidence of secondary vasculitis over 30 years is described as 5.1%, but in such cases, the question arises if the vasculitis is due to the anti-TNF α agents or simply a manifestation of the disease.¹⁶⁶

Eight patients with vasculitis associated with anti-TNF α therapy have been identified in a recent retrospective review of all patients treated with anti-TNF α therapy at the Mayo Clinic between 1st January 1998 and 31st March 2011.¹⁶⁷ The majority of cases ($n = 6$) were of female gender. Half of the patients had an original diagnosis of rheumatoid arthritis, while the other half had IBD. Five patients had been exposed to infliximab, 2 to etanercept and 1 to adalimumab. All cases were histologically proven to be small-vessel vasculitides. The mean duration for the development of vasculitis was 34.5 months, with most subjects having cutaneous involvement ($n = 5$) in the form of palpable purpura ($n = 4$). The peripheral nervous system was involved in half the cases and one case had renal involvement. The majority of cases ($n = 7$) improved after discontinuation of the drug and adjuvant therapy with a mean time to resolution of 6.9 months. None of the cases was re-challenged with anti-TNF α therapy, and none had recurred to follow-up.

Several hypotheses have been put forward to try and explain this association. Immune complexes such as anti-TNF α -TNF α complexes can deposit in the capillaries, leading to complement activation and a type III hypersensitivity reaction.¹⁶⁸ Another possible explanation would be the shift from a Th1 to Th2 response after anti-TNF α exposure leading to antibody-mediated immunity.¹⁶⁹

All these are plausible explanations, but as yet the association and pathogenesis of anti-TNF α -induced vasculitis are incompletely understood.

Granuloma annulare. Granuloma annulare is a benign asymptomatic papular eruption that can occur at all ages. It usually consists of grouped papules that enlarge in an annular fashion. It may have a skin-based colour or be erythematous. The prevalence of granuloma annulare may be high among patients with chronic disease,¹⁷⁰ but a rate of 3.9% is higher than what one would anticipate in an anti-TNF α -naïve rheumatology cohort.¹⁷¹ The precise aetiology of granuloma annulare is unknown, although TNF α inhibition results in activation of autoreactive T-cells, leading to autoimmune destruction of tissue¹⁶⁵ and development of this dermatological complication. Paradoxically, anti-TNF α agents have been unintentionally used to treat granuloma annulare in a complex case of CD.¹⁷²

Interstitial granulomatous dermatitis. Interstitial granulomatous dermatitis tends to present as a drug reaction or as dermatitis with arthritis. It presents as asymptomatic, annular, erythematous to violaceous plaques or linear cords with a typical distribution in the intertriginous areas, inner aspects of the thighs, arms and trunk.¹⁷³ It has been associated with multiple autoimmune disorders like rheumatoid arthritis, systemic lupus erythematosus, systemic vasculitides and lymphoproliferative disorders.^{174–178}

Histological examination reveals diffuse histiocytic and neutrophilic infiltrate, piecemeal fragmentation of the collagen and formation of small granulomas around degenerate areas. Palisaded neutrophilic and granulomatous dermatitis were the original histological description mentioned in the literature to describe the architectural distortion attributed to this dermatological manifestation.¹⁷⁹ The aetiopathology of this condition is still unclear. It had originally been postulated that immune complexes form in dermal blood vessels followed by activation of the complement system and subsequent neutrophilic infiltration leading to collagen degradation.¹⁷⁹

This cutaneous manifestation has been observed with anti-TNF α drugs^{180–182} and methotrexate.¹⁸³ After biologic drug exposure, it is postulated that interstitial granulomatous dermatitis is caused by antigen/antibody reactions leading to the formation of autoantibodies and histological inflammation. The immune-modulating effects of anti-TNF α drugs might alter the antigenicity of collagen, leading to an immune reaction or an impairment in collagen turnover.¹⁸⁴ All the cases

described in the literature are in rheumatological patients with none in the gastroenterology literature.^{180–182, 185} This might possibly be a reflection of the pharmacokinetic and pharmacodynamic interaction between the disease and drug, but also may be due to the granulomatous disposition of the patient phenotype.¹⁸⁵

Interstitial granulomatous dermatitis in a patient on anti-TNF α drugs is an indication to stop biological therapy. This usually results in prompt resolution of the rash.¹⁸⁰ Persistence of biological therapy while treating the patients with systemic steroids generally proves ineffectual,¹⁸⁶ apart from in one case report, where interstitial granulomatous dermatitis developed once the anti-TNF α drug was discontinued.¹⁸¹ TNF α is important in the induction and formation of granulomas.¹⁸⁵ Blocking this cytokine might impair granuloma formation in patients with a granulomatous diathesis. To this effect, paradoxically, a report of a patient with interstitial granulomatous dermatitis treated successfully with infliximab has been described in the literature.¹⁸⁷

Erythema nodosum. Erythema nodosum is an acute tender eruption of erythematous plaques and nodules, usually on the extensor surface of the lower limbs (Figure 5). Its aetiology is not completely understood, but it is described as a hypersensitivity reaction with



Figure 5 | Erythema nodosum. The erythematous nodule on the extensor surface of the lower limb can be seen in a Crohn's disease patient treated with anti-TNF biological therapy.

peri- and intravascular immune complex deposition in fatty and connective tissue. Pathological examination usually reveals septal inflammation and venulitis without evidence of vasculitis.¹⁸⁸ Possible differential diagnoses should be excluded.

Erythema nodosum is thought to be the commonest dermatological manifestation in IBD¹⁸⁹ and a prevalence of 2–15% has been quoted.^{190–193} Its incidence is higher in CD¹⁹³ and in the female sex, although the latter finding is still a contentious issue.¹⁹¹

Biological therapy has been used in the past to treat erythema nodosum in CD,¹⁹⁴ but reports in the last decade have described this dermatological manifestation as a possible complication of biological therapy, specifically for infliximab¹⁹⁵ and certolizumab pegol,¹⁹⁶ with no cases reported for adalimumab-treated patients in the IBD literature as yet. Infliximab-induced erythema nodosum has been described in ankylosing spondylitis^{197, 198} and Behcet's disease¹⁹⁹ patients as well. In all cases, the biological therapy had to be withheld and topical steroid treatment applied to heal this potential complication.

Cutaneous malignancies. The reported prevalence of nonmelanoma skin cancers in patients exposed to anti-TNF α agents ranges from 0.3% to 1.4%.^{113, 120, 153, 200} Anti-TNF therapy theoretically increases the incidence of malignancy^{113, 120} due to an alteration in immune surveillance, but earlier data were confounded by current and previous use of concomitant thiopurines or phototherapy.^{120, 153, 200}

In a recent retrospective cohort and nested case-control study, 26 403 patients with CD and 26 974 patients with UC were each matched to 3 non-IBD controls. The incidence rate ratio for NMSC was 1.64 (95% confidence interval 1.51–1.78). In the nested case-control study, 387 CD patients and 355 UC with NMSC patients were each matched to 4 IBD patients without NMSC. Recent (<90 days) and persistent (>365 days) biological monotherapy increases the risk of NMSC ~2-fold to a rate of 0.5 cases per 1000 person-years.³⁷

A meta-analysis looking at 74 randomised controlled trials using anti-TNF α therapy for at least 4 weeks has recently been conducted. The relative risk associated with all anti-TNF α monotherapies for all cancers was 0.99 (CI: 0.61–1.68), but this increased to 2.02 (CI: 1.11–3.95) when the incidence of NMSC was analysed. These recent data indicate that these types of cancer are a rare, but potential, complication of biological therapy.²⁰¹

The management of NMSC in anti-TNF α -exposed patients is mostly preventive. Patients should be referred

for baseline and routine skin checks and advice should be given regarding photoprotection. All newly diagnosed cancers should be referred for excision. Prompt cessation of the anti-TNF α drug is not routinely required.⁹⁴

A recent case report has described the occurrence of a cutaneous lymphoma in a 75-year-old gentleman with a 7-year history of UC after a short exposure to infliximab.²⁰² IBD alone is not associated with an increased risk of lymphoma, but previous and present anti-TNF α drug exposure has been associated with an increased risk of lymphoma. A SIR of 5.5 for past anti-TNF exposure (CI: 4.5–6.6) and 4.4 for current use (CI: 3.4–5.4) with or without thiopurine use has been recently described.²⁰³ These data apply to noncutaneous lymphoma. Data specific to this dermatological complication are not available in the literature.

Other rare dermatological manifestations. Dermatitis herpetiformis,⁹² pityriasis rosea,⁹² dermatitis sicca,⁹² pernio-like eruption,⁹⁶ interface dermatitis,¹⁵¹ bullous skin lesions, atypical of pemphigus vulgaris or bullous pemphigoid²⁰⁴ and, paradoxically, pyoderma gangrenosum²⁰⁵ have all been reported after exposure to anti-TNF α agents.

Thiopurine and anti-TNF α combination therapy

The use of corticosteroids, thiopurines and anti-TNF α therapy is associated with a 3- to 4-fold increased risk of acquiring an opportunistic infection. The risk is increased in patients receiving two or more of the above immune modulators and in patients over the age of 50 years.²⁰⁶ The high odds of opportunistic infections might just reflect trivial infections like oral candidiasis, but despite this, ongoing vigilance is advocated when more serious infections are endemic. As previously indicated, varicella zoster virus vaccination is advocated in non-exposed subjects prior to starting any immune modulators. This practice should be re-enforced in patients receiving combination therapy.

In an IBD nested case-control study, including 742 cases of NMSC and 2968 matched controls, the effect of combination therapy on the incidence of NMSC in comparison to thiopurine or anti-TNF α monotherapy has been quantified.³⁷ Patients with CD who recently used combination therapy had a 5.85-fold increased risk of developing NMSC when compared to the 3.56-fold risk in subjects on thiopurine monotherapy and 2.07-fold increased risk in subjects on anti-TNF α monotherapy.

For subjects having current therapy, the risk of developing NMSC was 4.27-fold for subjects on thiopurine

monotherapy, 2.18-fold for those on anti-TNF α monotherapy and 6.75-fold for those on combination therapy. These retrospective data suggest that immune suppression through thiopurine therapy poses an increased risk of NMSC when compared with the risk posed by anti-TNF α therapy. The effects of these two immune-suppressive therapies seem to be additive when combined.

Cutaneous lymphoma has been previously described in IBD subjects. In a retrospective study using an IBD and a cancer registry, the lymphoma incidence in the IBD cohort was calculated using the general population as a reference group.²⁰³ Present thiopurine usage gave a SIR of 1.4 (CI: 1.2–1.7) as compared to a SIR of 6.6 (4.4–8.8) for subjects on combination therapy of thiopurines and anti-TNF α therapy. In a prior meta-analysis, anti-TNF α monotherapy gave a SIR of 1.7 (CI: 0.5–7.1) when compared to subjects on thiopurine monotherapy.⁵⁰ The SIR was 3.23 (CI: 1.5–6.9) in subjects on combination therapy when compared to naïve patients. Once again, these data are not specific to cutaneous lymphomas. It is very difficult to dissect the separate risks imposed by anti-TNF α and thiopurine therapies, but these data indicate that combination therapy might give an additional lymphoma risk to subjects on combination therapy when compared with other subjects on respective monotherapies.

Natalizumab

Natalizumab is a monoclonal antibody targeting α_4 integrin. It is effective in the induction^{207, 208} and maintenance²⁰⁸ of remission in moderate-to-severe CD. Due to a potentially lethal adverse event of progressive multifocal leucoencephalopathy secondary to John Cunningham (JC) virus reactivation²⁰⁹ in 0.1% of patients,²¹⁰ the drug is only available in United States through restricted access programmes. Worldwide experience of natalizumab in CD and knowledge on potential dermatological side effects are limited. Allergic hypersensitivity reactions to natalizumab have been documented in up to 4% of patients, most commonly with the second infusion. Type 1 allergic reactions such as urticaria and allergic dermatitis were present in up to 46% and 7% of patients respectively (infusion-related hypersensitivity reactions during natalizumab treatment).²¹¹ Progressive serum-like sickness with swollen lips and urticarial erythema has also been documented in one patient following a second infusion of natalizumab, necessitating cessation of therapy.²¹² No significant dermatological safety signals have yet been reported with vedolizumab, a humanised monoclonal antibody to $\alpha 4\beta 7$ integrin.

Ustekinumab

Ustekinumab is a monoclonal antibody to the p40 subunit of IL12/23. Interleukin 12 and 23 have been implicated in the pathophysiology of CD. Interleukin 12, a heterodimer of p40 and p35 subunits, is essential in the differentiation of naïve CD4⁺ T-cells into Th1 T-cells, which predominantly produce interleukin 2 and interferon γ , which are pivotal in CD pathogenesis. Interleukin 23, a heterodimer of p40 and p19 subunits, is important in the differentiation of TH17 T-cells. Inhibiting both these cytokines has broad and diverse immunological effects. In a double-blind cross-over trial, ustekinumab has been shown to be effective in the treatment of moderate-to-severe CD.²¹³ More recent open-label use has supported this early data.²¹⁴

In IBD, ustekinumab has not shown any dermatological adverse manifestations, but its use has been limited. Ustekinumab has been recently approved for use in the United States, Canada and Europe for the treatment of moderate-to-severe psoriasis.²¹⁵ Paradoxically, four cases of worsening chronic plaque psoriasis after ustekinumab exposure with a change in morphology to a palmoplantar pustulosis phenotype have been described.^{216–219} In three cases, the patients had been exposed to anti-TNF α therapy prior to ustekinumab therapy.^{217–219}

The aetiology of this phenomenon is still as yet unclear, although thought to be similar to what is described after exposure to anti-TNF α therapy. Ustekinumab therapy may decrease IL-23 and TH17 cell-induced TNF α levels, leading to an elevation in IFN α in genetically predisposed subjects and subsequent development of psoriasis.

Accumulating evidence suggests the safety and efficacy of ustekinumab in psoriasis.^{220, 221} Despite the different mechanism of action on the IL-12 and IL-23 pathways, ustekinumab can be potentially used in the refractory anti-TNF α -induced psoriasis.

To date, cutaneous malignancies such as melanoma and nonmelanoma skin cancers have been observed in ustekinumab-exposed patients, but only to rates expected in the general population. The overall rate over 4 years of NMSC for ustekinumab is 0.61 (CI: 0.43–0.83) per 100 person-years, which is below that of 1.13 (CI: 0.14–4.09) per 100 person-years for the placebo groups in ustekinumab trials.^{220, 222} Despite reassuring data from three large randomised controlled trials (PHOENIX I,²¹⁵ PHOENIX 2²²³ and ACCEPT²²⁴), a recent case report has described two cases of rapid-onset multifocal squamous cell carcinomas in ustekinumab-exposed patients, thus perhaps indicating that baseline and routine skin

checks should still be a mainstay practice in ustekinumab-treated patients.²²²

Rates of infection were generally reported to be similar between patient groups exposed to different ustekinumab dosages (45 mg and 90 mg), apart from in the case of herpes zoster, where a 3-fold increase in incidence rates was noticed in subjects receiving the higher dose.²²⁰ In one case report, the occurrence of herpes zoster was thought to be opportunistic²²¹ and, in another case report, two subjects developed severe contiguous multidermatomal herpes zoster, 1 and 9 months after treatment initiation.²²⁵ The incidence of herpes zoster superinfection in ustekinumab-exposed patients is thought to be 2/37 patient-years with an incidence ratio of 53.5.²²⁶ It is generally considered that ustekinumab is a risk factor for the development of herpes zoster and appropriate vaccinations should be undertaken prior to treatment onset. Finally, one case report,²²⁷ described the onset of alopecia areata in a subject exposed to ustekinumab form psoriasis. In this instance, the psoriasis disease activity scores were improving, therefore raising the possibility that alopecia was a potential dermatological manifestation of ustekinumab. The pathophysiology of this reaction is as yet unexplained.

Tofacitinib

Tofacitinib is an oral Janus Kinase (JAK) inhibitor that has been investigated in both moderate-to-severe CD and UC. Tofacitinib inhibits JAK 1, 2 and 3 with *in vivo* specificity on JAK 1 and 2 over JAK 3. Altered signalling of JAK1 and 3 will lead to modulation of γ -chain-containing cytokines, predominantly interleukins 2, 4, 7, 9, 15 and 21. In a multicentre, double-blind, randomised, placebo-controlled study, tofacitinib showed promising results.²²⁸

The only dermatological adverse events observed with tofacitinib were acne in 13.2–16.8% of the cases and a nonspecific rash in up to 11.3% of exposed cases in clinical trials.^{229, 230}

CONCLUSION

Dermatological complications of IBD-related therapy are complex and diverse, spanning from local infections to autoimmune phenomena and malignancies. Gastroenterologists caring for IBD patients are regularly confronted with these side effects, with the increasing use of immunosuppressive drugs and biological therapies. A constant vigilance for potential complications is a necessity with prompt referrals organised when needed. Cutaneous infective complications should be diagnosed and treated early,

especially in patients at risk. Decisions to treat with immune suppression or combination therapy of anti-metabolites and antibodies to TNF α should take into consideration the age of the patients and the risk of infective complications. In the case of viral infections such as herpes simplex or varicella zoster, routine immunisation, serology check and patient vaccination prior to starting any immune suppressants should be standard practice.

Nonmelanoma skin cancers are an important entity, with an increased incidence now observed, possibly with the common use of combination immune-suppressive therapy. The highest incidence is in Caucasians over the age of 65 years. Patient education with appropriate sun protection advice and routine skin checks are of the utmost importance. Routine annual dermatological consultations are still uncommonly recommended in IBD patients. Prompt referral to dermatologists with a special interest in skin cancer should be standard practice. All lesions should be excised, and withdrawal of immune therapy is not compulsory, but should be discussed with the patient and a multidisciplinary team. Cutaneous lymphomas and malignant melanomas have been described, but the numbers are too small to draw any specific conclusions.

Knowing in advance which patients will get dermatological or systemic adverse events would be very clinically useful. Presently, no genetic or serological data are scientifically or commercially available that could be potentially used as a screening tool to identify subjects who could adversely react to different therapies. This is certainly an area where further research should be entertained.

More specific and targeted inhibitors and antibodies will be available in clinical use in the future. Vigilance for new cutaneous side effects should be maintained.

Dermatological adverse events of IBD-related therapy are diverse and potentially confused with extra-intestinal manifestations of the disease. The aetiology of some conditions is as yet unexplained. The management of IBD patients should still remain in the setting of a multidisciplinary team to ensure patient safety.

AUTHORSHIP

Guarantor of the article: Prof. Subrata Ghosh.

Author contributions: GWM, AWKL, JKB and SG conceived the study. GWM, AWKL and JKB did the literature search and drafted the manuscript. M-FD, YL, SMD, KN, GGK, MI, CS, LM, RP and SG critically reviewed the manuscript and approved the final draft. KN and LM provided the photographic data. GWM drafted the final manuscript. All authors approved the final version of the manuscript.

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